

CORE PURE / VECTORS, MATRICES & LINEAR TRANSFORMATIONS

Shortest Distances

Worked solutions

8 questions · 95 marks

Questions

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1. The plane Π has equation $3x + y + 2z = -4$. The point P has coordinates $(5, -5, 3)$.

(a) Calculate the shortest distance from P to Π .

[2]

The line L has vector equation

$$\mathbf{r} = \begin{pmatrix} 2 \\ 1 \\ -3 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -2 \\ 2 \end{pmatrix}$$

(b) Verify that P lies on L .

[2]

(c) Find the coordinates of the point where L meets Π .

[3]

(d) Determine the acute angle between L and Π .

[4]

(e) Use the results of parts (b), (c) and (d) to verify your answer to part (a).

[3]

Solution

(a) Using the Cartesian form of the point-to-plane distance formula,

$$\text{dist} = \frac{|ax_0 + by_0 + cz_0 - d|}{\sqrt{a^2 + b^2 + c^2}}$$

Here, the plane is

$$3x + y + 2z = -4$$

so $a = 3$, $b = 1$, $c = 2$, $d = -4$, and $P = (5, -5, 3)$.

Substituting into the formula:

$$\begin{aligned} \text{dist} &= \frac{|3(5) + 1(-5) + 2(3) - (-4)|}{\sqrt{3^2 + 1^2 + 2^2}} \\ &= \frac{|15 - 5 + 6 + 4|}{\sqrt{9 + 1 + 4}} \\ &= \frac{20}{\sqrt{14}} \\ &= \frac{10\sqrt{14}}{7} \end{aligned}$$

So the shortest distance from P to Π is $\frac{20}{\sqrt{14}} = \frac{10\sqrt{14}}{7}$.

(b) A general point on L has coordinates

$$(x, y, z) = (2 + \lambda, 1 - 2\lambda, -3 + 2\lambda)$$

For $P = (5, -5, 3)$ to lie on L , we need

$$2 + \lambda = 5$$

so

$$\lambda = 3$$

Now check the other coordinates:

$$\begin{aligned} y &= 1 - 2(3) = 1 - 6 = -5 \\ z &= -3 + 2(3) = -3 + 6 = 3 \end{aligned}$$

These match the coordinates of P .

Hence P lies on L , with parameter value $\lambda = 3$.

- (c) Let the point where L meets Π correspond to parameter λ .

From the line,

$$x = 2 + \lambda, \quad y = 1 - 2\lambda, \quad z = -3 + 2\lambda$$

Substitute into the plane equation $3x + y + 2z = -4$:

$$\begin{aligned} 3(2 + \lambda) + (1 - 2\lambda) + 2(-3 + 2\lambda) &= -4 \\ 6 + 3\lambda + 1 - 2\lambda - 6 + 4\lambda &= -4 \\ 1 + 5\lambda &= -4 \\ 5\lambda &= -5 \\ \lambda &= -1 \end{aligned}$$

Now substitute $\lambda = -1$ into the line:

$$\begin{aligned} x &= 2 + (-1) = 1 \\ y &= 1 - 2(-1) = 3 \\ z &= -3 + 2(-1) = -5 \end{aligned}$$

So the line meets the plane at $(1, 3, -5)$.

- (d) The direction vector of L is

$$\mathbf{d} = \begin{pmatrix} 1 \\ -2 \\ 2 \end{pmatrix}$$

and a normal vector to Π is

$$\mathbf{n} = \begin{pmatrix} 3 \\ 1 \\ 2 \end{pmatrix}$$

For the acute angle θ between a line and a plane,

$$\sin \theta = \frac{|\mathbf{d} \cdot \mathbf{n}|}{|\mathbf{d}| |\mathbf{n}|}$$

First find the dot product:

$$\begin{aligned} \mathbf{d} \cdot \mathbf{n} &= 1(3) + (-2)(1) + 2(2) \\ &= 3 - 2 + 4 \\ &= 5 \end{aligned}$$

Now find the magnitudes:

$$\begin{aligned} |\mathbf{d}| &= \sqrt{1^2 + (-2)^2 + 2^2} = \sqrt{1 + 4 + 4} = 3 \\ |\mathbf{n}| &= \sqrt{3^2 + 1^2 + 2^2} = \sqrt{9 + 1 + 4} = \sqrt{14} \end{aligned}$$

Therefore,

$$\sin \theta = \frac{5}{3\sqrt{14}}$$

So

$$\theta = \sin^{-1}\left(\frac{5}{3\sqrt{14}}\right) \approx 26.5^\circ$$

Hence the acute angle between L and Π is $\sin^{-1}\left(\frac{5}{3\sqrt{14}}\right) \approx 26.5^\circ$.

(e) From part (b), P corresponds to $\lambda = 3$ on the line.

From part (c), the point where L meets Π is $C(1, 3, -5)$, which corresponds to $\lambda = -1$.

So the change in parameter is

$$3 - (-1) = 4$$

Since the direction vector $\mathbf{d}$ of the line has magnitude

$$|\mathbf{d}| = 3$$

the length of the line segment from P to the plane along L is

$$PC = 4 \times 3 = 12$$

From part (d), the angle between the line and the plane is θ , where

$$\sin \theta = \frac{5}{3\sqrt{14}}$$

The perpendicular distance from P to the plane is the component of PC perpendicular to the plane:

$$\begin{aligned} \text{distance} &= PC \sin \theta \\ &= 12 \cdot \frac{5}{3\sqrt{14}} \\ &= \frac{60}{3\sqrt{14}} \\ &= \frac{20}{\sqrt{14}} \end{aligned}$$

This is exactly the same as the answer in part (a).

Hence the answer to part (a) is verified.

2. The points C and D have coordinates $(0, 0, 2)$ and $(3, 3, -1)$ respectively.

The planes P_1 and P_2 have equations $x + y + z = 3$ and $2x - y = 1$ respectively.

(a) Find the acute angle between the line CD and the plane P_1 . [4]

(b) Show that the line CD meets P_1 and P_2 in a common point, and find its coordinates. [5]

(c) (i) Find $(\mathbf{i} + \mathbf{j} + \mathbf{k}) \times (2\mathbf{i} - \mathbf{j})$. [1]

(ii) Hence find the acute angle between the planes P_1 and P_2 . [3]

(iii) Find the shortest distance between the point C and the line of intersection of the planes P_1 and P_2 . [4]

Solution

(a) The direction vector of the line CD is

$$\overrightarrow{CD} = \begin{pmatrix} 3-0 \\ 3-0 \\ -1-2 \end{pmatrix} = \begin{pmatrix} 3 \\ 3 \\ -3 \end{pmatrix}$$

So we may use the parallel direction vector

$$\mathbf{d} = \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}$$

For the plane $P_1 : x + y + z = 3$, a normal vector is

$$\mathbf{n}_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

If α is the acute angle between a line and a plane, then

$$\sin \alpha = \frac{|\mathbf{d} \cdot \mathbf{n}_1|}{|\mathbf{d}| |\mathbf{n}_1|}$$

Now

$$\mathbf{d} \cdot \mathbf{n}_1 = (1)(1) + (1)(1) + (-1)(1) = 1$$

$$|\mathbf{d}| = \sqrt{1^2 + 1^2 + (-1)^2} = \sqrt{3}$$

$$|\mathbf{n}_1| = \sqrt{1^2 + 1^2 + 1^2} = \sqrt{3}$$

Hence

$$\sin \alpha = \frac{1}{\sqrt{3}\sqrt{3}} = \frac{1}{3}$$

So

$$\alpha = \sin^{-1}\left(\frac{1}{3}\right) \approx 19.5^\circ$$

The acute angle between CD and P_1 is $\sin^{-1}(1/3) \approx 19.5^\circ$.

(b) The line CD through $C = (0, 0, 2)$ with direction vector $\mathbf{d}$ has equation

$$\mathbf{r} = \begin{pmatrix} 0 \\ 0 \\ 2 \end{pmatrix} + t \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}$$

So a general point on CD is

$$(x, y, z) = (t, t, 2 - t)$$

First, substitute into $P_1 : x + y + z = 3$:

$$t + t + (2 - t) = 3$$

$$t + 2 = 3$$

$$t = 1$$

Now substitute into $P_2 : 2x - y = 1$:

$$2t - t = 1$$

$$t = 1$$

Both planes are met when $t = 1$, so the line CD meets P_1 and P_2 at the same point.

When $t = 1$,

$$(x, y, z) = (1, 1, 2 - 1) = (1, 1, 1)$$

Therefore the common point is $(1, 1, 1)$.

(c) (i)

$$(\mathbf{i} + \mathbf{j} + \mathbf{k}) \times (2\mathbf{i} - \mathbf{j}) = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 1 & 1 \\ 2 & -1 & 0 \end{vmatrix}$$

Expanding,

$$= \mathbf{i}(1 \cdot 0 - 1 \cdot (-1)) - \mathbf{j}(1 \cdot 0 - 1 \cdot 2) + \mathbf{k}(1 \cdot (-1) - 1 \cdot 2)$$

$$= \mathbf{i}(1) - \mathbf{j}(-2) + \mathbf{k}(-3)$$

$$= \mathbf{i} + 2\mathbf{j} - 3\mathbf{k}$$

Hence

$$(\mathbf{i} + \mathbf{j} + \mathbf{k}) \times (2\mathbf{i} - \mathbf{j}) = \mathbf{i} + 2\mathbf{j} - 3\mathbf{k}$$

(ii) The acute angle between two planes is the acute angle between their normal vectors.

For

$$P_1 : x + y + z = 3 \quad \text{and} \quad P_2 : 2x - y = 1$$

normal vectors are

$$\mathbf{n}_1 = \mathbf{i} + \mathbf{j} + \mathbf{k}, \quad \mathbf{n}_2 = 2\mathbf{i} - \mathbf{j}$$

If θ is the acute angle between the planes, then since $|\mathbf{n}_1 \times \mathbf{n}_2| = |\mathbf{n}_1| |\mathbf{n}_2| \sin \theta$,

$$\sin \theta = \frac{|\mathbf{n}_1 \times \mathbf{n}_2|}{|\mathbf{n}_1| |\mathbf{n}_2|}$$

From part (i), $\mathbf{n}_1 \times \mathbf{n}_2 = \mathbf{i} + 2\mathbf{j} - 3\mathbf{k}$, so

$$|\mathbf{n}_1 \times \mathbf{n}_2| = \sqrt{1^2 + 2^2 + (-3)^2} = \sqrt{14}$$

$$|\mathbf{n}_1| = \sqrt{1^2 + 1^2 + 1^2} = \sqrt{3}$$

$$|\mathbf{n}_2| = \sqrt{2^2 + (-1)^2 + 0^2} = \sqrt{5}$$

So

$$\sin \theta = \frac{\sqrt{14}}{\sqrt{3}\sqrt{5}} = \sqrt{\frac{14}{15}}$$

Hence

$$\theta = \sin^{-1} \left(\sqrt{\frac{14}{15}} \right) \approx 75.0^\circ$$

The acute angle between the planes is $\sin^{-1} \sqrt{14/15} \approx 75.0^\circ$.

(iii) From part (b), the planes intersect at the point

$$A = (1, 1, 1)$$

From part (i), a direction vector for their line of intersection is

$$\mathbf{v} = \begin{pmatrix} 1 \\ 2 \\ -3 \end{pmatrix}$$

So the line of intersection is

$$\ell : \mathbf{r} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 2 \\ -3 \end{pmatrix}$$

We need the shortest distance from $C = (0, 0, 2)$ to ℓ .

Using the formula

$$\text{dist} = \frac{|(\mathbf{p} - \mathbf{a}) \times \mathbf{b}|}{|\mathbf{b}|}$$

with

$$\mathbf{p} = \begin{pmatrix} 0 \\ 0 \\ 2 \end{pmatrix}, \quad \mathbf{a} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 1 \\ 2 \\ -3 \end{pmatrix}$$

we have

$$\mathbf{p} - \mathbf{a} = \begin{pmatrix} 0 \\ 0 \\ 2 \end{pmatrix} - \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ -1 \\ 1 \end{pmatrix}$$

Now find the cross product:

$$(\mathbf{p} - \mathbf{a}) \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -1 & -1 & 1 \\ 1 & 2 & -3 \end{vmatrix}$$

So

$$\begin{aligned} (\mathbf{p} - \mathbf{a}) \times \mathbf{b} &= \mathbf{i}((-1)(-3) - 1 \cdot 2) - \mathbf{j}((-1)(-3) - 1 \cdot 1) + \mathbf{k}((-1) \cdot 2 - (-1) \cdot 1) \\ &= \mathbf{i}(1) - \mathbf{j}(2) + \mathbf{k}(-1) \\ &= \mathbf{i} - 2\mathbf{j} - \mathbf{k} \end{aligned}$$

Hence

$$\begin{aligned} |(\mathbf{p} - \mathbf{a}) \times \mathbf{b}| &= \sqrt{1^2 + (-2)^2 + (-1)^2} = \sqrt{6} \\ |\mathbf{b}| &= \sqrt{1^2 + 2^2 + (-3)^2} = \sqrt{14} \end{aligned}$$

Therefore

$$\text{dist} = \frac{\sqrt{6}}{\sqrt{14}} = \sqrt{\frac{3}{7}}$$

So the shortest distance is

$$\sqrt{\frac{3}{7}} \approx 0.655$$

3. The plane Π has equation

$$\mathbf{r} = \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} + \mu \begin{pmatrix} 4 \\ -1 \\ -2 \end{pmatrix}$$

where λ and μ are scalar parameters.

(a) Show that vector $\mathbf{i} - 2\mathbf{j} + 3\mathbf{k}$ is perpendicular to Π . [2]

(b) Hence find a Cartesian equation of Π . [2]

The line ℓ has equation

$$\mathbf{r} = \begin{pmatrix} -2 \\ 1 \\ 4 \end{pmatrix} + t \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

where t is a scalar parameter.

The point A lies on ℓ . Given that the shortest distance between A and Π is $\frac{9}{\sqrt{14}}$

(c) determine the possible coordinates of A . [4]

Solution

(a) Let

$$\mathbf{n} = \mathbf{i} - 2\mathbf{j} + 3\mathbf{k} = \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix}$$

The plane Π has direction vectors

$$\begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 4 \\ -1 \\ -2 \end{pmatrix}$$

Check the dot products:

$$\mathbf{n} \cdot \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} = 1(1) + (-2)(2) + 3(1) = 1 - 4 + 3 = 0$$

$$\mathbf{n} \cdot \begin{pmatrix} 4 \\ -1 \\ -2 \end{pmatrix} = 1(4) + (-2)(-1) + 3(-2) = 4 + 2 - 6 = 0$$

Since $\mathbf{n}$ is perpendicular to both direction vectors of the plane, $\mathbf{n}$ is perpendicular to Π .

Therefore $\mathbf{i} - 2\mathbf{j} + 3\mathbf{k}$ is perpendicular to Π .

(b) Using the normal vector $\mathbf{n} = \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix}$ and the point $\begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix}$ on Π , the Cartesian equation is

$$\mathbf{n} \cdot \left(\begin{pmatrix} x \\ y \\ z \end{pmatrix} - \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix} \right) = 0$$

So

$$\begin{aligned}\begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix} \cdot \begin{pmatrix} x-2 \\ y+1 \\ z-3 \end{pmatrix} &= 0 \\ (x-2) - 2(y+1) + 3(z-3) &= 0 \\ x-2-2y-2+3z-9 &= 0 \\ x-2y+3z-13 &= 0\end{aligned}$$

Therefore a Cartesian equation of Π is

$$x - 2y + 3z = 13$$

(c) Since A lies on the line ℓ ,

$$\mathbf{r} = \begin{pmatrix} -2 \\ 1 \\ 4 \end{pmatrix} + t \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

the coordinates of A are

$$A = (-2 + t, 1 + t, 4 + t)$$

From part (b), the plane is

$$x - 2y + 3z = 13$$

Using the point-to-plane distance formula

$$\text{dist} = \frac{|ax_0 + by_0 + cz_0 - d|}{\sqrt{a^2 + b^2 + c^2}}$$

with $a = 1$, $b = -2$, $c = 3$, $d = 13$, we get

$$\begin{aligned}\frac{|(-2+t) - 2(1+t) + 3(4+t) - 13|}{\sqrt{1^2 + (-2)^2 + 3^2}} &= \frac{9}{\sqrt{14}} \\ \frac{|-2+t-2-2t+12+3t-13|}{\sqrt{14}} &= \frac{9}{\sqrt{14}} \\ \frac{|2t-5|}{\sqrt{14}} &= \frac{9}{\sqrt{14}}\end{aligned}$$

Hence

$$|2t - 5| = 9$$

So

$$2t - 5 = 9 \quad \text{or} \quad 2t - 5 = -9$$

giving

$$t = 7 \quad \text{or} \quad t = -2$$

Substitute back into the line:

$$\begin{aligned}t = 7 &\implies A = (5, 8, 11) \\ t = -2 &\implies A = (-4, -1, 2)\end{aligned}$$

Therefore the possible coordinates of A are $(5, 8, 11)$ or $(-4, -1, 2)$.

4. The line ℓ_1 has vector equation

$$\mathbf{r} = \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$$

The line ℓ_2 is the line of intersection of the planes

$$x - y + z = 6$$

and

$$2x + y - z = 3$$

(i) Find the shortest distance between ℓ_1 and ℓ_2 . [5]

(ii) Find a Cartesian equation of the plane which contains ℓ_1 and is parallel to ℓ_2 . [2]

Solution

(i) First find a vector equation for ℓ_2 .

Let $z = t$. Then the plane equations become

$$\begin{aligned} x - y + t &= 6 \\ 2x + y - t &= 3 \end{aligned}$$

Adding these equations gives

$$3x = 9$$

so

$$x = 3$$

Substituting into $x - y + t = 6$,

$$\begin{aligned} 3 - y + t &= 6 \\ -y &= 3 - t \\ y &= t - 3 \end{aligned}$$

Hence ℓ_2 has vector equation

$$\mathbf{r} = \begin{pmatrix} 3 \\ -3 \\ 0 \end{pmatrix} + t \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$$

So we can take

$$\mathbf{a}_1 = \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix}, \quad \mathbf{d}_1 = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}, \quad \mathbf{a}_2 = \begin{pmatrix} 3 \\ -3 \\ 0 \end{pmatrix}, \quad \mathbf{d}_2 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$$

For the shortest distance between two skew lines,

$$\text{dist} = \frac{|(\mathbf{a}_2 - \mathbf{a}_1) \cdot (\mathbf{d}_1 \times \mathbf{d}_2)|}{|\mathbf{d}_1 \times \mathbf{d}_2|}$$

First calculate the cross product:

$$\begin{aligned}\mathbf{d}_1 \times \mathbf{d}_2 &= \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} \times \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \\ &= \begin{pmatrix} 2(1) - (-1)(1) \\ -(1(1) - (-1)(0)) \\ 1(1) - 2(0) \end{pmatrix} \\ &= \begin{pmatrix} 3 \\ -1 \\ 1 \end{pmatrix}\end{aligned}$$

Also,

$$\mathbf{a}_2 - \mathbf{a}_1 = \begin{pmatrix} 3 \\ -3 \\ 0 \end{pmatrix} - \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix} = \begin{pmatrix} 1 \\ -2 \\ -3 \end{pmatrix}$$

Now find the scalar triple product:

$$\begin{aligned}(\mathbf{a}_2 - \mathbf{a}_1) \cdot (\mathbf{d}_1 \times \mathbf{d}_2) &= \begin{pmatrix} 1 \\ -2 \\ -3 \end{pmatrix} \cdot \begin{pmatrix} 3 \\ -1 \\ 1 \end{pmatrix} \\ &= 1(3) + (-2)(-1) + (-3)(1) \\ &= 3 + 2 - 3 \\ &= 2\end{aligned}$$

And

$$\begin{aligned}|\mathbf{d}_1 \times \mathbf{d}_2| &= \sqrt{3^2 + (-1)^2 + 1^2} \\ &= \sqrt{11}\end{aligned}$$

Therefore

$$\text{dist} = \frac{|2|}{\sqrt{11}} = \frac{2}{\sqrt{11}}$$

The shortest distance between ℓ_1 and ℓ_2 is $\frac{2}{\sqrt{11}}$.

(ii) A plane containing ℓ_1 and parallel to ℓ_2 must contain both direction vectors

$$\mathbf{d}_1 = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}, \quad \mathbf{d}_2 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$$

So a normal vector to the plane is

$$\mathbf{n} = \mathbf{d}_1 \times \mathbf{d}_2 = \begin{pmatrix} 3 \\ -1 \\ 1 \end{pmatrix}$$

The plane passes through the point $(2, -1, 3)$ on ℓ_1 . Hence its equation is

$$\begin{aligned}\mathbf{n} \cdot \left(\begin{pmatrix} x \\ y \\ z \end{pmatrix} - \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix} \right) &= 0 \\ \begin{pmatrix} 3 \\ -1 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} x-2 \\ y+1 \\ z-3 \end{pmatrix} &= 0 \\ 3(x-2) - (y+1) + (z-3) &= 0\end{aligned}$$

Simplifying,

$$3x - 6 - y - 1 + z - 3 = 0$$

$$3x - y + z - 10 = 0$$

So a Cartesian equation of the plane is

$$3x - y + z = 10$$

5. (a) Show that the three planes with equations

$$\begin{aligned}x + \lambda y + z &= 4 \\2x - y + 2z &= 3 \\2x + y - z &= 5\end{aligned}\quad [3]$$

where λ is a constant, meet at a unique point except for one value of λ , which is to be determined.

- (b) In the case $\lambda = 2$, use matrices to find the point of intersection P of the planes, showing your method clearly. [3]

The line ℓ has equation

$$\frac{x+1}{1} = \frac{y-1}{-1} = \frac{z}{1}$$

- (c) Find a vector equation of ℓ . [2]
- (d) Find the shortest distance between the point P and ℓ . [4]
- (e) (i) Show that ℓ is parallel to the plane $2x + y - z = 5$. [3]
(ii) Find the distance between ℓ and the plane $2x + y - z = 5$. [2]

Solution

- (a) Let the coefficient matrix be

$$A = \begin{pmatrix} 1 & \lambda & 1 \\ 2 & -1 & 2 \\ 2 & 1 & -1 \end{pmatrix}$$

The three planes meet at a unique point if and only if the simultaneous equations have a unique solution, that is, if and only if $\det A \neq 0$.

Expanding along the first row,

$$\begin{aligned}\det A &= 1 \begin{vmatrix} -1 & 2 \\ 1 & -1 \end{vmatrix} - \lambda \begin{vmatrix} 2 & 2 \\ 2 & -1 \end{vmatrix} + 1 \begin{vmatrix} 2 & -1 \\ 2 & 1 \end{vmatrix} \\ &= 1((-1)(-1) - 2(1)) - \lambda(2(-1) - 2(2)) + (2(1) - (-1)(2)) \\ &= (1 - 2) - \lambda(-2 - 4) + (2 + 2) \\ &= -1 + 6\lambda + 4 \\ &= 6\lambda + 3 \\ &= 3(2\lambda + 1)\end{aligned}$$

So the planes meet at a unique point whenever

$$3(2\lambda + 1) \neq 0$$

that is,

$$\lambda \neq -\frac{1}{2}$$

Hence the exceptional value is

$$\lambda = -\frac{1}{2}$$

For $\lambda = -\frac{1}{2}$, the first plane is

$$x - \frac{1}{2}y + z = 4$$

and multiplying by 2 gives

$$2x - y + 2z = 8$$

but the second plane is

$$2x - y + 2z = 3$$

so the system is inconsistent. Therefore there is no common point in this case.

Hence the three planes meet at a unique point for all λ except $\lambda = -\frac{1}{2}$.

(b) When $\lambda = 2$, the equations are

$$x + 2y + z = 4$$

$$2x - y + 2z = 3$$

$$2x + y - z = 5$$

Write this system as

$$A\mathbf{x} = \mathbf{b}$$

where

$$A = \begin{pmatrix} 1 & 2 & 1 \\ 2 & -1 & 2 \\ 2 & 1 & -1 \end{pmatrix}, \quad \mathbf{x} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 4 \\ 3 \\ 5 \end{pmatrix}$$

Since $\det A \neq 0$, the matrix is invertible, so

$$\mathbf{x} = A^{-1}\mathbf{b}$$

Using a calculator to find the inverse,

$$A^{-1} = \begin{pmatrix} -\frac{1}{15} & \frac{1}{5} & \frac{1}{3} \\ \frac{2}{5} & -\frac{1}{5} & 0 \\ \frac{4}{15} & \frac{1}{5} & -\frac{1}{3} \end{pmatrix}$$

Hence

$$\mathbf{x} = \begin{pmatrix} -\frac{1}{15} & \frac{1}{5} & \frac{1}{3} \\ \frac{2}{5} & -\frac{1}{5} & 0 \\ \frac{4}{15} & \frac{1}{5} & -\frac{1}{3} \end{pmatrix} \begin{pmatrix} 4 \\ 3 \\ 5 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}$$

So the point of intersection is

$$P = (2, 1, 0)$$

(c) Let the common parameter be t . Then

$$\frac{x+1}{1} = \frac{y-1}{-1} = \frac{z}{1} = t$$

So

$$x + 1 = t, \quad y - 1 = -t, \quad z = t$$

hence

$$x = t - 1, \quad y = 1 - t, \quad z = t$$

Therefore a vector equation of ℓ is

$$\mathbf{r} = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} + t \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}$$

(d) From part (b), $P = (2, 1, 0)$.

Take the point

$$Q = (-1, 1, 0)$$

on the line ℓ , and the direction vector of ℓ as

$$\mathbf{d} = \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}$$

Then

$$\overrightarrow{QP} = \overrightarrow{OP} - \overrightarrow{OQ} = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} - \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 3 \\ 0 \\ 0 \end{pmatrix}$$

Using

$$\text{dist} = \frac{|\overrightarrow{QP} \times \mathbf{d}|}{|\mathbf{d}|}$$

First find the cross product:

$$\begin{aligned} \overrightarrow{QP} \times \mathbf{d} &= \begin{pmatrix} 3 \\ 0 \\ 0 \end{pmatrix} \times \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} \\ &= \begin{pmatrix} 0 \\ -3 \\ -3 \end{pmatrix} \end{aligned}$$

So

$$\begin{aligned} |\overrightarrow{QP} \times \mathbf{d}| &= \sqrt{0^2 + (-3)^2 + (-3)^2} \\ &= \sqrt{18} \\ &= 3\sqrt{2} \end{aligned}$$

Also,

$$|\mathbf{d}| = \sqrt{1^2 + (-1)^2 + 1^2} = \sqrt{3}$$

Hence

$$\begin{aligned} \text{dist} &= \frac{3\sqrt{2}}{\sqrt{3}} \\ &= \sqrt{6} \end{aligned}$$

Therefore the shortest distance between P and ℓ is

$$\sqrt{6}$$

(e) (i) The plane

$$2x + y - z = 5$$

has normal vector

$$\mathbf{n} = \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix}$$

The line ℓ has direction vector

$$\mathbf{d} = \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}$$

Now

$$\begin{aligned}\mathbf{d} \cdot \mathbf{n} &= \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix} \\ &= 1(2) + (-1)(1) + 1(-1) \\ &= 2 - 1 - 1 \\ &= 0\end{aligned}$$

So ℓ is either parallel to the plane or lies in the plane.

Take the point $Q = (-1, 1, 0)$ on ℓ . Substituting into the plane equation,

$$2(-1) + 1 - 0 = -1$$

which is not equal to 5.

So Q is not on the plane, hence ℓ does not lie in the plane.

Therefore ℓ is parallel to the plane $2x + y - z = 5$.

- (ii) Since ℓ is parallel to the plane, the distance from ℓ to the plane is the distance from any point on ℓ to the plane.

Using $Q = (-1, 1, 0)$ and the formula

$$\text{dist} = \frac{|ax_0 + by_0 + cz_0 - d|}{\sqrt{a^2 + b^2 + c^2}}$$

for the plane $2x + y - z = 5$,

$$\begin{aligned}\text{dist} &= \frac{|2(-1) + 1(1) + (-1)(0) - 5|}{\sqrt{2^2 + 1^2 + (-1)^2}} \\ &= \frac{|-2 + 1 - 5|}{\sqrt{6}} \\ &= \frac{6}{\sqrt{6}} \\ &= \sqrt{6}\end{aligned}$$

Therefore the distance between ℓ and the plane is

$$\sqrt{6}$$

6. (a) The plane Π passes through the points $A(1, -2, 3)$, $B(2, 0, 3)$ and $C(2, -2, 2)$.

The point D has coordinates $(4, 1, 0)$. Find the shortest distance between D and Π .

[4]

- (b) Lines ℓ_1 and ℓ_2 have the following equations.

$$\ell_1 : \mathbf{r} = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix}$$

$$\ell_2 : \mathbf{r} = \begin{pmatrix} 3 \\ -1 \\ 4 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}$$

Find, in exact form, the distance between ℓ_1 and ℓ_2 .

[5]

Solution

- (a) First find two direction vectors in the plane:

$$\overrightarrow{AB} = \begin{pmatrix} 2-1 \\ 0-(-2) \\ 3-3 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}, \quad \overrightarrow{AC} = \begin{pmatrix} 2-1 \\ -2-(-2) \\ 2-3 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$$

A normal vector to the plane is

$$\begin{aligned} \mathbf{n} &= \overrightarrow{AB} \times \overrightarrow{AC} \\ &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 2 & 0 \\ 1 & 0 & -1 \end{vmatrix} \\ &= \mathbf{i}(2 \cdot (-1) - 0 \cdot 0) - \mathbf{j}(1 \cdot (-1) - 0 \cdot 1) + \mathbf{k}(1 \cdot 0 - 2 \cdot 1) \\ &= \begin{pmatrix} -2 \\ 1 \\ -2 \end{pmatrix} \end{aligned}$$

So we may take the normal as

$$\mathbf{n} = \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix}$$

The plane has equation

$$2x - y + 2z = d$$

Since it passes through $A(1, -2, 3)$,

$$\begin{aligned} d &= 2(1) - (-2) + 2(3) \\ &= 2 + 2 + 6 \\ &= 10 \end{aligned}$$

Hence the plane is

$$\Pi : 2x - y + 2z = 10$$

Using the point-to-plane distance formula with $D(4, 1, 0)$,

$$\begin{aligned}\text{dist} &= \frac{|2(4) - 1 + 2(0) - 10|}{\sqrt{2^2 + (-1)^2 + 2^2}} \\ &= \frac{|8 - 1 - 10|}{\sqrt{4 + 1 + 4}} \\ &= \frac{3}{3} \\ &= 1\end{aligned}$$

The shortest distance between D and Π is 1.

(b) Write the lines as

$$\begin{aligned}\ell_1 : \mathbf{r} &= \mathbf{a}_1 + \lambda \mathbf{b}_1, & \mathbf{a}_1 &= \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}, & \mathbf{b}_1 &= \begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix} \\ \ell_2 : \mathbf{r} &= \mathbf{a}_2 + \mu \mathbf{b}_2, & \mathbf{a}_2 &= \begin{pmatrix} 3 \\ -1 \\ 4 \end{pmatrix}, & \mathbf{b}_2 &= \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}\end{aligned}$$

For the distance between two skew lines, use

$$\text{dist} = \frac{|(\mathbf{a}_2 - \mathbf{a}_1) \cdot (\mathbf{b}_1 \times \mathbf{b}_2)|}{|\mathbf{b}_1 \times \mathbf{b}_2|}$$

First find the common normal direction:

$$\begin{aligned}\mathbf{b}_1 \times \mathbf{b}_2 &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & 1 & 2 \\ 1 & -2 & 1 \end{vmatrix} \\ &= \mathbf{i}(1 \cdot 1 - 2 \cdot (-2)) - \mathbf{j}(2 \cdot 1 - 2 \cdot 1) + \mathbf{k}(2 \cdot (-2) - 1 \cdot 1) \\ &= \begin{pmatrix} 5 \\ 0 \\ -5 \end{pmatrix}\end{aligned}$$

Also,

$$\mathbf{a}_2 - \mathbf{a}_1 = \begin{pmatrix} 3 - 1 \\ -1 - 2 \\ 4 - (-1) \end{pmatrix} = \begin{pmatrix} 2 \\ -3 \\ 5 \end{pmatrix}$$

Now calculate the numerator:

$$\begin{aligned}(\mathbf{a}_2 - \mathbf{a}_1) \cdot (\mathbf{b}_1 \times \mathbf{b}_2) &= \begin{pmatrix} 2 \\ -3 \\ 5 \end{pmatrix} \cdot \begin{pmatrix} 5 \\ 0 \\ -5 \end{pmatrix} \\ &= 2(5) + (-3)(0) + 5(-5) \\ &= 10 - 25 \\ &= -15\end{aligned}$$

So

$$|(\mathbf{a}_2 - \mathbf{a}_1) \cdot (\mathbf{b}_1 \times \mathbf{b}_2)| = 15$$

And the denominator is

$$\begin{aligned}|\mathbf{b}_1 \times \mathbf{b}_2| &= \sqrt{5^2 + 0^2 + (-5)^2} \\ &= \sqrt{25 + 25} \\ &= 5\sqrt{2}\end{aligned}$$

Therefore,

$$\begin{aligned}\text{dist} &= \frac{15}{5\sqrt{2}} \\ &= \frac{3}{\sqrt{2}} \\ &= \frac{3\sqrt{2}}{2}\end{aligned}$$

The distance between ℓ_1 and ℓ_2 is $\frac{3\sqrt{2}}{2}$.

7. The position vectors of the points A, B, C, D are

$$\mathbf{i} + 2\mathbf{j}, \quad 2\mathbf{i} + 2\mathbf{j}, \quad \mathbf{i} + 3\mathbf{j} + 2\mathbf{k}, \quad 2\mathbf{i} + 3\mathbf{j} + m\mathbf{k}$$

respectively, where m is an integer.

It is given that the shortest distance between the line through A and D and the line through B and C is $\frac{1}{\sqrt{14}}$.

(a) Show that the only possible value of m is 1. [7]

(b) Find the shortest distance of C from the line through A and D . [3]

(c) Show that the acute angle between the planes ABD and ACD is $\cos^{-1}\left(\frac{\sqrt{3}}{2}\right)$. [4]

Solution

(a) Writing the points in coordinates,

$$A = (1, 2, 0), \quad B = (2, 2, 0), \quad C = (1, 3, 2), \quad D = (2, 3, m)$$

So the direction vectors of the two lines are

$$\overrightarrow{AD} = \begin{pmatrix} 1 \\ 1 \\ m \end{pmatrix}, \quad \overrightarrow{BC} = \begin{pmatrix} -1 \\ 1 \\ 2 \end{pmatrix}$$

Also,

$$\overrightarrow{AB} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

Using the formula for the distance between two skew lines,

$$\text{dist} = \frac{|\overrightarrow{AB} \cdot (\overrightarrow{AD} \times \overrightarrow{BC})|}{|\overrightarrow{AD} \times \overrightarrow{BC}|}$$

First find the cross product:

$$\begin{aligned} \overrightarrow{AD} \times \overrightarrow{BC} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 1 & m \\ -1 & 1 & 2 \end{vmatrix} \\ &= \mathbf{i}(1 \cdot 2 - m \cdot 1) - \mathbf{j}(1 \cdot 2 - m(-1)) + \mathbf{k}(1 \cdot 1 - 1(-1)) \\ &= (2 - m)\mathbf{i} - (2 + m)\mathbf{j} + 2\mathbf{k} \end{aligned}$$

Now the scalar triple product is

$$\begin{aligned} \overrightarrow{AB} \cdot (\overrightarrow{AD} \times \overrightarrow{BC}) &= \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 2 - m \\ -2 - m \\ 2 \end{pmatrix} \\ &= 2 - m \end{aligned}$$

Next,

$$\begin{aligned} |\overrightarrow{AD} \times \overrightarrow{BC}|^2 &= (2 - m)^2 + (-2 - m)^2 + 2^2 \\ &= (m^2 - 4m + 4) + (m^2 + 4m + 4) + 4 \\ &= 2m^2 + 12 \end{aligned}$$

So

$$|\overrightarrow{AD} \times \overrightarrow{BC}| = \sqrt{2m^2 + 12}$$

Therefore the shortest distance is

$$\frac{|2 - m|}{\sqrt{2m^2 + 12}}$$

This is given to be $\frac{1}{\sqrt{14}}$, so

$$\frac{|2 - m|}{\sqrt{2m^2 + 12}} = \frac{1}{\sqrt{14}}$$

Squaring both sides,

$$\begin{aligned} 14(2 - m)^2 &= 2m^2 + 12 \\ 14(m^2 - 4m + 4) &= 2m^2 + 12 \\ 14m^2 - 56m + 56 &= 2m^2 + 12 \\ 12m^2 - 56m + 44 &= 0 \\ 3m^2 - 14m + 11 &= 0 \end{aligned}$$

Factorising,

$$3m^2 - 14m + 11 = (3m - 11)(m - 1)$$

So

$$(3m - 11)(m - 1) = 0$$

Hence

$$m = \frac{11}{3} \quad \text{or} \quad m = 1$$

Since m is an integer, the only possible value is

$$m = 1$$

(b) Using $m = 1$,

$$D = (2, 3, 1)$$

For the line through A and D ,

$$\overrightarrow{AD} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

Also,

$$\overrightarrow{AC} = \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix}$$

Using the point-to-line distance formula,

$$\text{dist} = \frac{|\overrightarrow{AC} \times \overrightarrow{AD}|}{|\overrightarrow{AD}|}$$

Now

$$\begin{aligned} \overrightarrow{AC} \times \overrightarrow{AD} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & 1 & 2 \\ 1 & 1 & 1 \end{vmatrix} \\ &= \mathbf{i}(1 \cdot 1 - 2 \cdot 1) - \mathbf{j}(0 \cdot 1 - 2 \cdot 1) + \mathbf{k}(0 \cdot 1 - 1 \cdot 1) \\ &= -\mathbf{i} + 2\mathbf{j} - \mathbf{k} \end{aligned}$$

So

$$\left| \overrightarrow{AC} \times \overrightarrow{AD} \right| = \sqrt{(-1)^2 + 2^2 + (-1)^2} = \sqrt{6}$$

and

$$\left| \overrightarrow{AD} \right| = \sqrt{1^2 + 1^2 + 1^2} = \sqrt{3}$$

Hence the shortest distance is

$$\frac{\sqrt{6}}{\sqrt{3}} = \sqrt{2}$$

So the shortest distance of C from the line through A and D is

$$\sqrt{2}$$

(c) With $m = 1$,

$$\overrightarrow{AB} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad \overrightarrow{AC} = \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix}, \quad \overrightarrow{AD} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

A normal to plane ABD is

$$\begin{aligned} \mathbf{n}_{ABD} &= \overrightarrow{AB} \times \overrightarrow{AD} \\ &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 0 & 0 \\ 1 & 1 & 1 \end{vmatrix} \\ &= 0\mathbf{i} - \mathbf{j} + \mathbf{k} \end{aligned}$$

A normal to plane ACD is

$$\begin{aligned} \mathbf{n}_{ACD} &= \overrightarrow{AC} \times \overrightarrow{AD} \\ &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & 1 & 2 \\ 1 & 1 & 1 \end{vmatrix} \\ &= -\mathbf{i} + 2\mathbf{j} - \mathbf{k} \end{aligned}$$

Using the acute angle formula for two planes,

$$\cos \theta = \frac{|\mathbf{n}_{ABD} \cdot \mathbf{n}_{ACD}|}{|\mathbf{n}_{ABD}| |\mathbf{n}_{ACD}|}$$

Now

$$\begin{aligned} \mathbf{n}_{ABD} \cdot \mathbf{n}_{ACD} &= \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} -1 \\ 2 \\ -1 \end{pmatrix} \\ &= 0 - 2 - 1 \\ &= -3 \end{aligned}$$

Hence

$$|\mathbf{n}_{ABD} \cdot \mathbf{n}_{ACD}| = 3$$

Also,

$$|\mathbf{n}_{ABD}| = \sqrt{0^2 + (-1)^2 + 1^2} = \sqrt{2}$$

and

$$|\mathbf{n}_{ACD}| = \sqrt{(-1)^2 + 2^2 + (-1)^2} = \sqrt{6}$$

Therefore

$$\begin{aligned}\cos \theta &= \frac{3}{\sqrt{2} \sqrt{6}} \\ &= \frac{3}{\sqrt{12}} \\ &= \frac{3}{2\sqrt{3}} \\ &= \frac{\sqrt{3}}{2}\end{aligned}$$

So the acute angle between the planes ABD and ACD is

$$\theta = \cos^{-1} \left(\frac{\sqrt{3}}{2} \right)$$

That is,

$$\theta = \frac{\pi}{6} = 30^\circ$$

8. (a) Given that $\mathbf{u} = \lambda\mathbf{i} - 2\mathbf{j} + 2\mathbf{k}$ and $\mathbf{v} = \mathbf{i} + \mathbf{j} - \mathbf{k}$, find the following, giving your answers in terms of λ .

(i) $\mathbf{u} \cdot \mathbf{v}$ [1]

(ii) $\mathbf{u} \times \mathbf{v}$ [2]

(b) Hence determine

(i) the acute angle between the planes $2x - 2y + 2z = 11$ and $x + y - z = 7$ [3]

(ii) the shortest distance between the lines

$$\mathbf{r} = \mathbf{i} - \mathbf{k} + s(\mathbf{i} - 2\mathbf{j} + 2\mathbf{k})$$

and

$$\mathbf{r} = 2\mathbf{i} + 3\mathbf{j} + 3\mathbf{k} + t(\mathbf{i} + \mathbf{j} - \mathbf{k})$$

giving your answer as a multiple of $\sqrt{2}$.

[3]

Solution

(a) (i)

$$\mathbf{u} \cdot \mathbf{v} = (\lambda)(1) + (-2)(1) + (2)(-1) = \lambda - 2 - 2 = \lambda - 4$$

(ii)

$$\begin{aligned}\mathbf{u} \times \mathbf{v} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \lambda & -2 & 2 \\ 1 & 1 & -1 \end{vmatrix} \\ &= \mathbf{i}((-2)(-1) - 2(1)) - \mathbf{j}(\lambda(-1) - 2(1)) + \mathbf{k}(\lambda(1) - (-2)(1)) \\ &= \mathbf{i}(2 - 2) - \mathbf{j}(-\lambda - 2) + \mathbf{k}(\lambda + 2) \\ &= 0\mathbf{i} + (\lambda + 2)\mathbf{j} + (\lambda + 2)\mathbf{k} \\ &= (\lambda + 2)(\mathbf{j} + \mathbf{k})\end{aligned}$$

(b) (i) The acute angle between two planes is the acute angle between their normal vectors.

For the planes

$$2x - 2y + 2z = 11 \quad \text{and} \quad x + y - z = 7$$

the normal vectors are

$$\mathbf{n}_1 = \begin{pmatrix} 2 \\ -2 \\ 2 \end{pmatrix}, \quad \mathbf{n}_2 = \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}$$

Using part (a)(i), $\mathbf{n}_1 \cdot \mathbf{n}_2$ is $\mathbf{u} \cdot \mathbf{v}$ with $\lambda = 2$, so

$$\mathbf{n}_1 \cdot \mathbf{n}_2 = 2 - 4 = -2$$

Also

$$|\mathbf{n}_1| = \sqrt{2^2 + (-2)^2 + 2^2} = \sqrt{12} = 2\sqrt{3}$$

$$|\mathbf{n}_2| = \sqrt{1^2 + 1^2 + (-1)^2} = \sqrt{3}$$

Using

$$\cos \theta = \frac{|\mathbf{n}_1 \cdot \mathbf{n}_2|}{|\mathbf{n}_1| |\mathbf{n}_2|}$$

gives

$$\begin{aligned}\cos \theta &= \frac{|-2|}{(2\sqrt{3})(\sqrt{3})} \\ &= \frac{2}{6} \\ &= \frac{1}{3}\end{aligned}$$

Therefore

$$\theta = \cos^{-1} \left(\frac{1}{3} \right) \approx 70.5^\circ$$

Hence the acute angle between the planes is $\cos^{-1} \left(\frac{1}{3} \right) \approx 70.5^\circ$.

(ii) Write the lines as

$$L_1 : \mathbf{r} = \mathbf{a}_1 + s\mathbf{b}_1, \quad L_2 : \mathbf{r} = \mathbf{a}_2 + t\mathbf{b}_2$$

where

$$\mathbf{a}_1 = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}, \quad \mathbf{b}_1 = \begin{pmatrix} 1 \\ -2 \\ 2 \end{pmatrix}, \quad \mathbf{a}_2 = \begin{pmatrix} 2 \\ 3 \\ 3 \end{pmatrix}, \quad \mathbf{b}_2 = \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}$$

Using the formula for the distance between two skew lines,

$$\text{dist} = \frac{|(\mathbf{a}_2 - \mathbf{a}_1) \cdot (\mathbf{b}_1 \times \mathbf{b}_2)|}{|\mathbf{b}_1 \times \mathbf{b}_2|}$$

From part (a)(ii), with $\lambda = 1$,

$$\mathbf{b}_1 \times \mathbf{b}_2 = (1+2)(\mathbf{j} + \mathbf{k}) = 3(\mathbf{j} + \mathbf{k}) = \begin{pmatrix} 0 \\ 3 \\ 3 \end{pmatrix}$$

Also

$$\mathbf{a}_2 - \mathbf{a}_1 = \begin{pmatrix} 2-1 \\ 3-0 \\ 3-(-1) \end{pmatrix} = \begin{pmatrix} 1 \\ 3 \\ 4 \end{pmatrix}$$

Now

$$\begin{aligned} (\mathbf{a}_2 - \mathbf{a}_1) \cdot (\mathbf{b}_1 \times \mathbf{b}_2) &= \begin{pmatrix} 1 \\ 3 \\ 4 \end{pmatrix} \cdot \begin{pmatrix} 0 \\ 3 \\ 3 \end{pmatrix} \\ &= 1 \cdot 0 + 3 \cdot 3 + 4 \cdot 3 \\ &= 21 \end{aligned}$$

and

$$|\mathbf{b}_1 \times \mathbf{b}_2| = \sqrt{0^2 + 3^2 + 3^2} = \sqrt{18} = 3\sqrt{2}$$

Therefore

$$\begin{aligned} \text{dist} &= \frac{21}{3\sqrt{2}} \\ &= \frac{7}{\sqrt{2}} \\ &= \frac{7\sqrt{2}}{2} \end{aligned}$$

Hence the shortest distance is

$$\frac{7\sqrt{2}}{2}$$